# Process FMEA - Bracket Welding Cell - Section 1: Executive Summary & Scope



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: Highest RPN failure mode is inconsistent weld penetration from electrode wear. Secondary control 1.2: Mitigation adds an automated electrode wear check every 500 cycles.

- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Mitigation adds an automated electrode wear check every 500 cycles.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |

# Process FMEA - Bracket Welding Cell - Section 2: Architecture & Process Flow

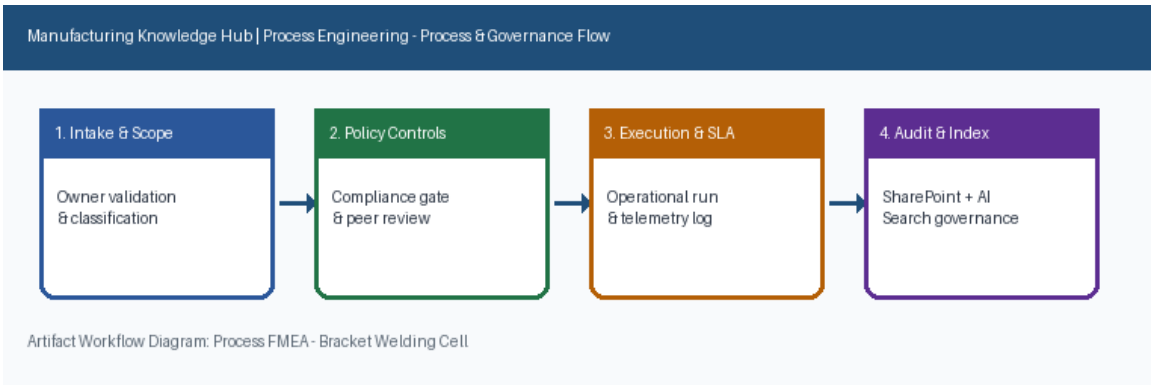


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Mitigation adds an automated electrode wear check every 500 cycles. Secondary control 2.2: Detection controls were upgraded from manual visual check to inline vision inspection.

- Mitigation adds an automated electrode wear check every 500 cycles.
- Detection controls were upgraded from manual visual check to inline vision inspection.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-2 | Carlos Ibarra | 3 business days | Stage 2 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 3: Governance Standards & Controls



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Detection controls were upgraded from manual visual check to inline vision inspection. Secondary control 3.2: FMEA is reviewed and updated after any process or equipment change.

- Detection controls were upgraded from manual visual check to inline vision inspection.
- FMEA is reviewed and updated after any process or equipment change.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-3 | Carlos Ibarra | 10 business days | Stage 3 Verified |

# Process FMEA - Bracket Welding Cell - Section 4: Operational KPIs & Benchmarks

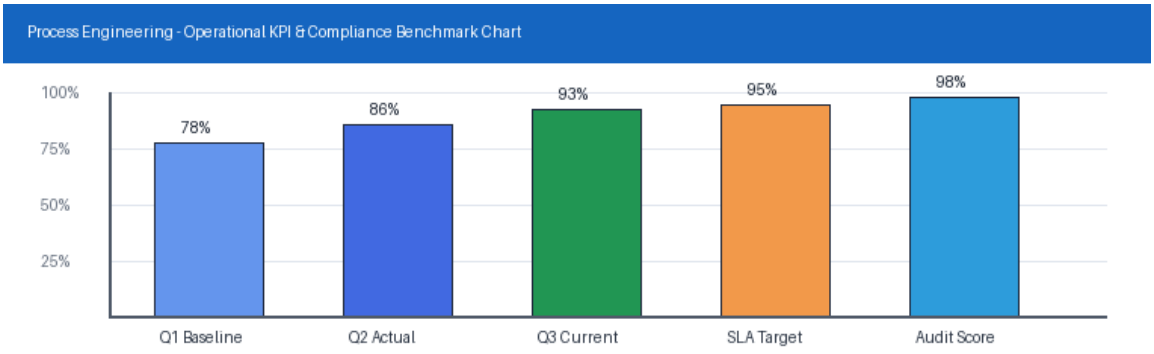


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 4 (Operational KPIs & Benchmarks) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Tracks baseline performance, target SLAs, and quarterly benchmark progression.

Operational requirement 4.1: FMEA is reviewed and updated after any process or equipment change. Secondary control 4.2: Highest RPN failure mode is inconsistent weld penetration from electrode wear.

- FMEA is reviewed and updated after any process or equipment change.
- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-4 | Carlos Ibarra | 5 business days | Stage 4 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 5: Roles, Responsibilities & RACI



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 5 (Roles, Responsibilities & RACI) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Clarifies ownership across accountable executives, operational leads, and reviewers.

Operational requirement 5.1: Highest RPN failure mode is inconsistent weld penetration from electrode wear. Secondary control 5.2: Mitigation adds an automated electrode wear check every 500 cycles.

- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Mitigation adds an automated electrode wear check every 500 cycles.
- Section 5 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-5 | Carlos Ibarra | 3 business days | Stage 5 Verified |

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 6: Standard Operating Procedures

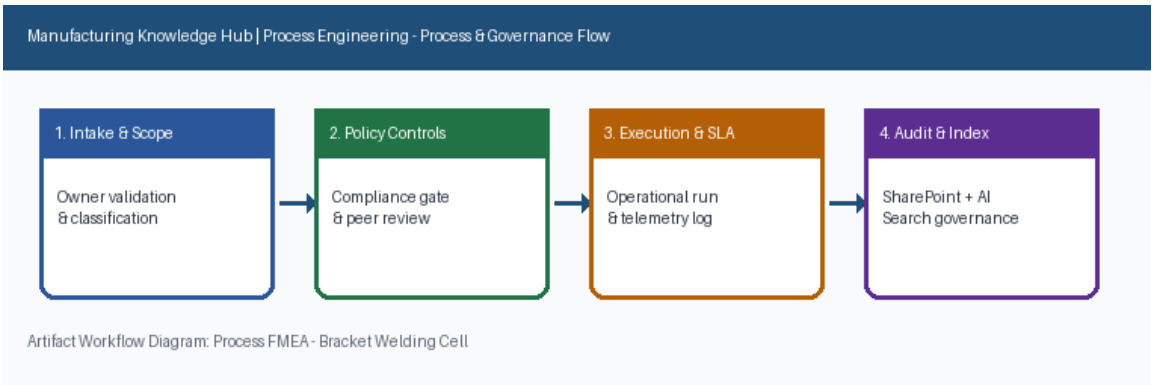


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 6 (Standard Operating Procedures) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Provides step-by-step execution checklists, intake criteria, and verification gates.

Operational requirement 6.1: Mitigation adds an automated electrode wear check every 500 cycles. Secondary control 6.2: Detection controls were upgraded from manual visual check to inline vision inspection.

- Mitigation adds an automated electrode wear check every 500 cycles.
- Detection controls were upgraded from manual visual check to inline vision inspection.
- Section 6 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-6 | Carlos Ibarra | 10 business days | Stage 6 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 7: Risk Register & Mitigations



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 7 (Risk Register & Mitigations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Catalogs operational, compliance, and delivery risks alongside preventive mitigations.

Operational requirement 7.1: Detection controls were upgraded from manual visual check to inline vision inspection. Secondary control 7.2: FMEA is reviewed and updated after any process or equipment change.

- Detection controls were upgraded from manual visual check to inline vision inspection.
- FMEA is reviewed and updated after any process or equipment change.
- Section 7 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-7 | Carlos Ibarra | 5 business days | Stage 7 Verified |

# Process FMEA - Bracket Welding Cell - Section 8: Quality Assurance & Audits

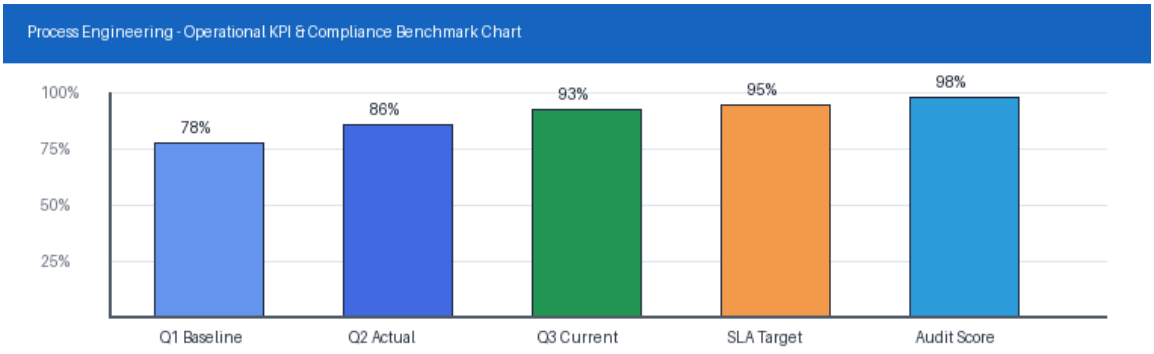


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 8 (Quality Assurance & Audits) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Specifies sampling frequencies, inspection tolerances, and nonconformance workflows.

Operational requirement 8.1: FMEA is reviewed and updated after any process or equipment change. Secondary control 8.2: Highest RPN failure mode is inconsistent weld penetration from electrode wear.

- FMEA is reviewed and updated after any process or equipment change.
- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Section 8 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-8 | Carlos Ibarra | 3 business days | Stage 8 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 9: Exception Handling & Escalations

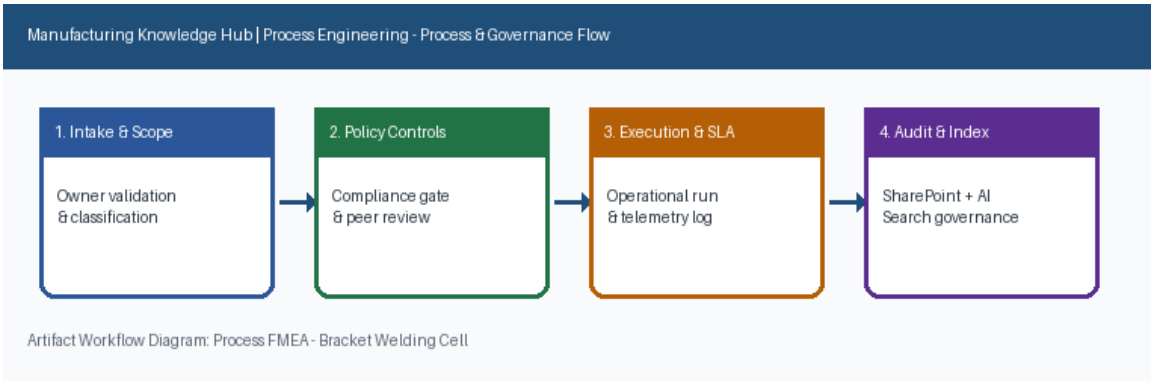


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 9 (Exception Handling & Escalations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Outlines severity tiers, response SLAs, and cross-functional escalation paths.

Operational requirement 9.1: Highest RPN failure mode is inconsistent weld penetration from electrode wear. Secondary control 9.2: Mitigation adds an automated electrode wear check every 500 cycles.

- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Mitigation adds an automated electrode wear check every 500 cycles.
- Section 9 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-9 | Carlos Ibarra | 10 business days | Stage 9 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 10: Cross-Functional Dependencies



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 10 (Cross-Functional Dependencies) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Documents upstream and downstream team dependencies and service-level commitments.

Operational requirement 10.1: Mitigation adds an automated electrode wear check every 500 cycles. Secondary control 10.2: Detection controls were upgraded from manual visual check to inline vision inspection.

- Mitigation adds an automated electrode wear check every 500 cycles.
- Detection controls were upgraded from manual visual check to inline vision inspection.
- Section 10 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-10 | Carlos Ibarra | 5 business days | Stage 10 Verified |

# Process FMEA - Bracket Welding Cell - Section 11: Continuous Improvement Roadmap

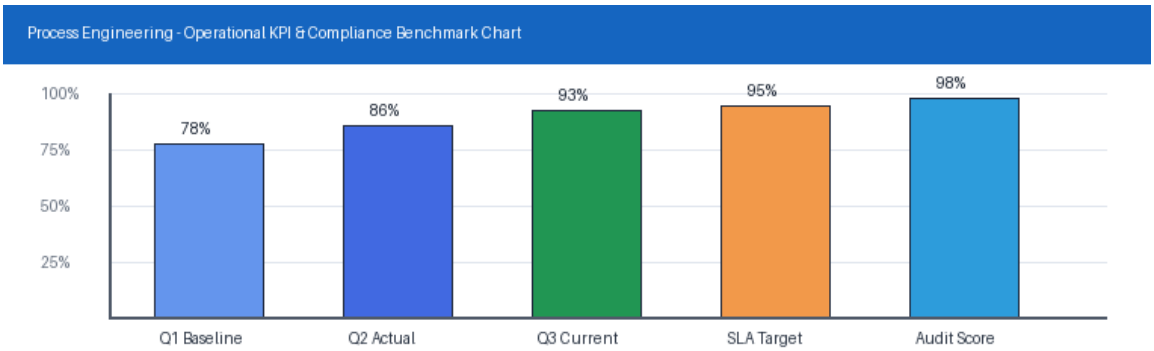


Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 11 (Continuous Improvement Roadmap) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Schedules quarterly optimization milestones, automation targets, and capability reviews.

Operational requirement 11.1: Detection controls were upgraded from manual visual check to inline vision inspection. Secondary control 11.2: FMEA is reviewed and updated after any process or equipment change.

- Detection controls were upgraded from manual visual check to inline vision inspection.
- FMEA is reviewed and updated after any process or equipment change.
- Section 11 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-11 | Carlos Ibarra | 3 business days | Stage 11 Verified |




Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

# Process FMEA - Bracket Welding Cell - Section 12: Appendix, Lineage & Sign-off



Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning. Section 12 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 12.1: FMEA is reviewed and updated after any process or equipment change. Secondary control 12.2: Highest RPN failure mode is inconsistent weld penetration from electrode wear.

- FMEA is reviewed and updated after any process or equipment change.
- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Section 12 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

| Control Area | Owner | Target SLA | Verification Status |
|---|---|---|---|
| Process Engineering | Carlos Ibarra | 5 business days | Verified |
| Process FMEA - Bracket Weldi Gate 1 | Carlos Ibarra | 3 business days | Active |
| Process FMEA - Bracket Weldi Gate 2 | Carlos Ibarra | 10 business days | Monitored |
| Sec-12 | Carlos Ibarra | 10 business days | Stage 12 Verified |

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online